

Jindong Cao

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Education

Carnegie Mellon University, College of Engineering
M.S. in Electrical and Computer Engineering, GPA: 3.75/4.0

Pittsburgh, USA
Aug 2023 – Dec 2024

Sun Yat-sen University, School of Computer Science and Engineering
B.S. in Computer Science and Technology, GPA: 3.9/4.0

Guangzhou, China
Aug 2019 – Jul 2023

Experience

TikTok
Machine Learning Engineer

San Jose, USA
Jan 2025 – Present

- Led migration of core push systems to a unified C++ scheduling framework, decoupling business logic from global decision-making and driving meaningful user growth.
- Owned end-to-end push modeling pipeline, including feature engineering, sample/label construction, model design, and online integration with ranking and frequency control.
- Built an end-to-end generative recommendation pipeline and evaluation benchmark to improve cold-start efficiency, measure generation diversity, better infer user interests, and drive meaningful business impact.
- Leveraged LLMs and VLMs to construct semantic IDs, optimize vLLM-based inference, and introduce cross-domain and cross-modal modeling for generating user-interest candidate categories.

Amazon
Software Development Engineer Intern

San Diego, USA
May 2024 – Aug 2024

- Built internal full-stack tools with React, Java, and TypeScript, improving service accessibility and operational efficiency.

ZMO.AI
Machine Learning Engineer Intern

Shenzhen, China
Mar 2023 – May 2023

- Developed an AI image editing pipeline in Python integrating ChatGPT, SAM, GroundingDINO, and Stable Diffusion.
- Applied pruning to BLIP-2 and CLIP, reducing memory usage by 30% and runtime by 40%.

Huawei Technologies
Software Development Engineer Intern

Dongguan, China
Jul 2022 – Sep 2022

- Refactored a data center query system in C++, reducing average query latency by 70%.

Research

Weakly supervised prompt learning for low-resource medical image classification, Pattern Recognition
Co-author *Jan 2024*

- Leveraged transferable representations from large-scale pre-trained vision-language models to address low-resource medical image classification for clinical decision support.
- Improved zero-shot and few-shot classification performance via additional prompt-embedding structures.

Skills

Python, C++, SQL, PyTorch, TensorFlow, Deep Learning, Machine Learning, Recommendation System